Functional & Non-Functional Requirement

Opening Question

If we are developing a Library Management System, what should the system do?

- Login
- Search books
- Borrow books
- Return books
- Reserve books
- Calculate fines
- Generate reports

What is a Functional Requirement?

A **Functional Requirement (FR)** specifies a **service, function, or behavior that a software system must provide**.

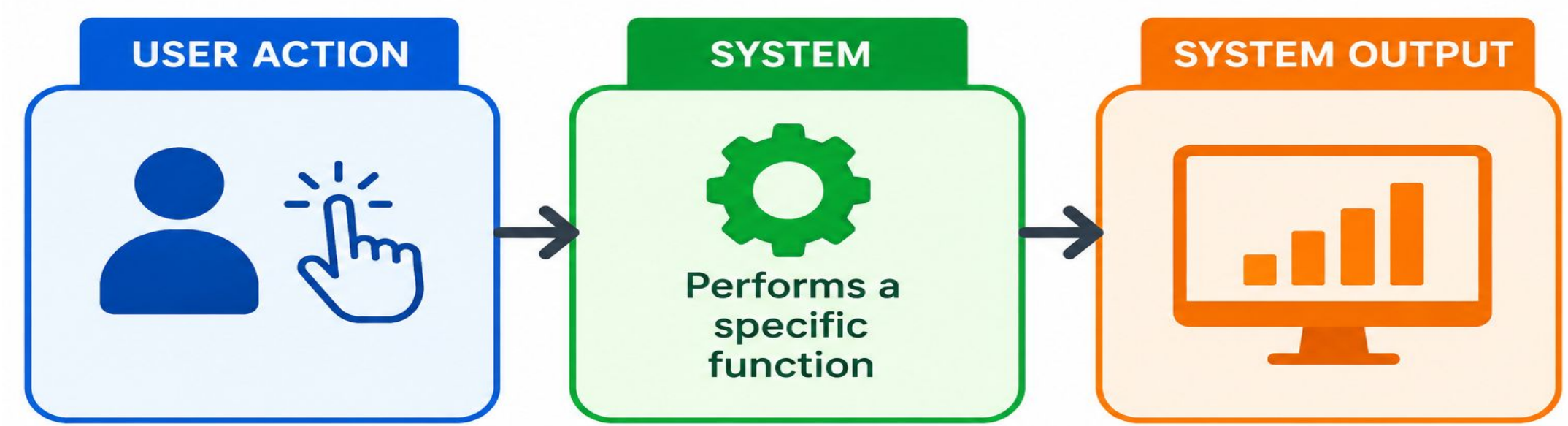
Functional requirements describe WHAT the system should do

Examples:

- The system shall allow students to log in.
- The system shall allow students to search for books.
- The system shall allow librarians to add new books.
- The system shall calculate overdue fines.

Functional Requirement

Functional Requirement = System Behavior



Functional Requirement

Example

Student: Searches for "Software Engineering"

↓

System: Searches the database

↓

System: Displays matching books

This behavior can be represented as a functional requirement:

FR-01: The system shall allow students to search for books by title, author, or ISBN.

Who Performs the Function?

An actor is a person, organization, or external system that interacts with the software.

For our Library Management System:

- Actors
- Student
- Librarian
- Administrator
- Payment System

Actor	Functions
Student	Search, borrow, return, reserve books
Librarian	Add/update/delete books, issue books, generate reports
Administrator	Manage users and system configuration
Payment System	Process online fine payments

Functional Requirements by Actor

Student

- Login
- Search Book
- View Availability
- Borrow Book
- Return Book
- Reserve Book
- View Borrowing History

Librarian

- Login
- Add Book
- Update Book
- Remove Book
- Issue Book
- Receive Book
- Generate Report

How to Write a Functional Requirement

Use a standard pattern:

The system shall + action + object + condition

Example: The system shall allow students to search books by title.

Component	Example
Subject	The system
Mandatory keyword	shall
Action	allow
Actor	students
Function	search
Object	books
Condition/criteria	by title

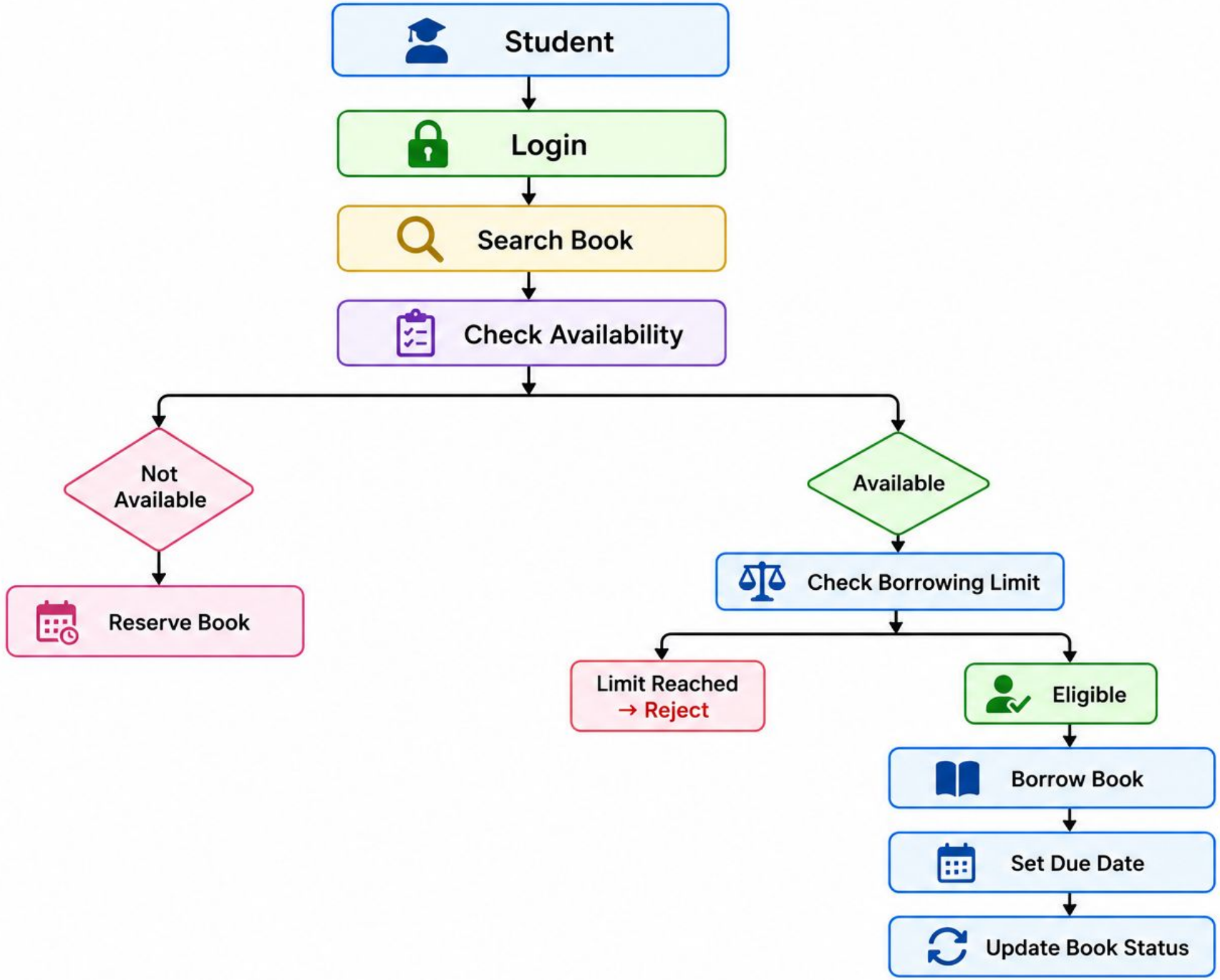
Functional Requirements Hierarchy

High-Level Requirement

FR-04: The system shall support book borrowing.

More Detailed Requirements

- **FR-04.1:** The system shall verify student eligibility.
- **FR-04.2:** The system shall verify book availability.
- **FR-04.3:** The system shall record the borrowing transaction.
- **FR-04.4:** The system shall assign a due date.
- **FR-04.5:** The system shall update the book status to "Borrowed."


Student
Login
Search Book
Check Availability
Not Available
Reserve Book
Available
Check Borrowing Limit
Limit Reached → Reject
Eligible
Borrow Book
Set Due Date
Update Book Status

Classroom Exercise

Online Food Delivery System

A university wants to develop an Online Food Delivery System that allows students and staff to order food from campus restaurants.

i. Identify the actors
ii. Write **10 functional requirements**.

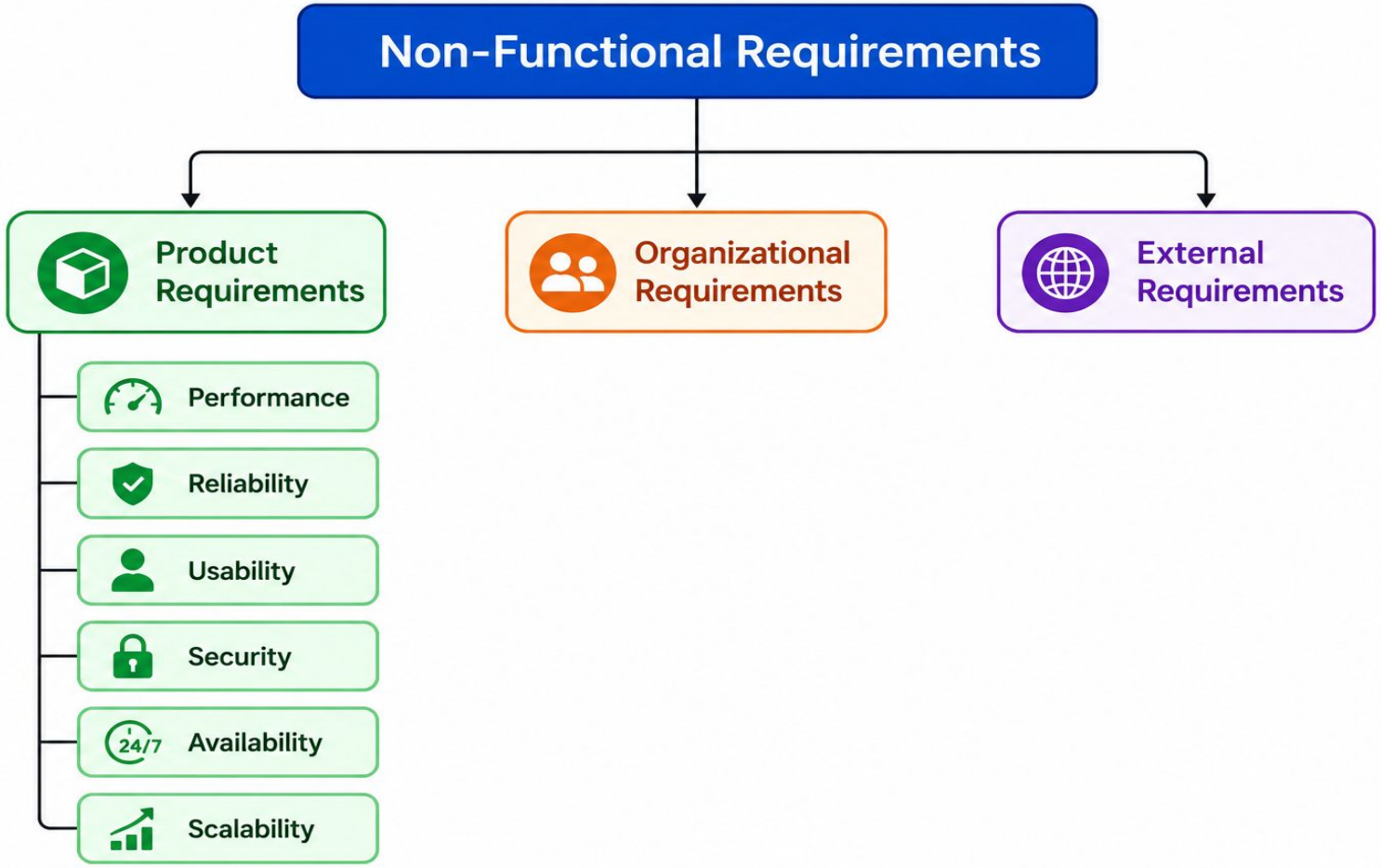
What is a Non-Functional Requirement?

A Non-Functional Requirement (NFR) specifies quality attributes, constraints, or conditions under which a software system must operate.

Functional requirements describe WHAT the system does. **Non-functional requirements describe HOW WELL the system performs or what constraints it must satisfy.**

Functional Requirement: *What should the system do?*
Non-Functional Requirement: *How well should it do it, and what constraints must it satisfy?*

Major Categories of Non-Functional Requirements



Product Requirements

Category	Description	Poor Example	Good Example
Performance	Specifies how quickly or efficiently the system should respond.	The system shall be fast.	**NFR-01:** The system shall display book search results within **2 seconds** under normal operating conditions.
Reliability	Specifies how reliably the system should operate without failure.	The system shall be reliable.	**NFR-02:** The system shall successfully process at least **99.5% of valid book-borrowing transactions** without system failure.
Availability	Specifies how much time the system should remain operational and accessible.	The system shall always be available.	**NFR-03:** The Library Management System shall be available **99.9% of the time**, excluding scheduled maintenance.
Usability	Specifies how easy and effective the system should be for users.	The system shall be user-friendly.	**NFR-04:** At least **90% of first-time student users** shall be able to complete a book search without assistance.
Security	Specifies requirements for protecting data, users, and system access.	The system shall be secure.	**NFR-05:** The system shall lock a user account after **5 consecutive unsuccessful login attempts**.
Scalability	Specifies the system's ability to handle increasing numbers of users, transactions, or data.	The system shall support many users.	**NFR-06:** The system shall support at least **10,000 registered users** without degradation of core functionality.

How to Write Good Non-Functional Requirements

Quality Attribute	Bad / Subjective Requirement	Problem	Better / Measurable Requirement
Security	The system shall be secure.	"Secure" is vague and does not specify how security will be measured.	The system shall lock a user account after 5 consecutive failed login attempts.
Performance	The system shall be very fast.	"Very fast" is subjective and cannot be objectively tested.	The system shall display search results within 2 seconds.
Usability	The system shall be easy to use.	"Easy to use" means different things to different users.	At least 90% of first-time users shall successfully complete a book search without assistance.

Key Principle: A good NFR should be specific, measurable, and testable whenever possible.

Avoid: *fast, easy, secure, reliable, user-friendly, efficient*

Prefer: measurable values such as *2 seconds, 99.9% availability, 5 attempts, 90% task completion,* etc.

Quality Attribute → Measurement

Quality Attribute	Possible Measurement
Performance	Response time
Availability	% uptime
Reliability	Failure rate
Usability	Task completion rate
Security	Number of failed attempts / security controls
Scalability	Maximum concurrent users
Maintainability	Time required to modify a module
Recovery	Recovery time / recovery point

Functional + Non-Functional Together

- **Requirement 1: Function**
  - **FR-02:** The system shall allow students to search books by title, author, ISBN, and category.
- **Requirement 2: Performance**
  - **NFR-01:** The system shall return search results within **2 seconds**.
- **Requirement 3: Availability**
  - **NFR-02:** The search service shall be available **99.9% of the time**.
- **Requirement 4: Security**
  - **NFR-03:** Only authenticated users shall access personalized borrowing information.

Together, these requirements define both the function and the quality constraints surrounding that function.

Functional vs. Non-Functional Requirements

Functional Requirement	Non-Functional Requirement
Describes **what** the system does	Describes **how well** the system performs
Defines system functions	Defines quality attributes/constraints
Usually expressed as system behavior	Usually expressed as measurable quality constraints
Login	Response time
Search books	Availability
Borrow books	Security
Return books	Usability
Calculate fine	Reliability

Why NFR is important in SRE?

Classroom Exercise

- Identify whether they are **Functional (FR)** or **Non-Functional (NFR)**.

Requirement
The system shall allow students to borrow books.
Search results shall appear within 2 seconds.
The system shall allow librarians to add books.
The system shall be available 99.9% of the time.
The system shall calculate overdue fines.
The system shall lock an account after 5 failed login attempts.
The system shall support 10,000 users.
The system shall generate monthly reports.

